Special Series, Convergence and Divergence Tests for Series

SERIES/TEST	SERIES	CONVERGENCE OR DIVERGENCE	COMMENTS
Geometric Series	$\sum_{n=1}^{\infty} ar^{n-1}$, $a \neq 0$	• Converges if $\lvert r\rvert < 1$ and its sum is $s = \dfrac{a}{1-r}$, that is, $\sum_{n=1}^{\infty} ar^{n-1} = \dfrac{a}{1-r}$ • Diverges if $\lvert r\rvert \geq 1$.	Useful for comparison tests if the nth term an of a series is similar to ar^{n-1}. Note that a is the first term of the series.
Test for Divergence	$\sum_{n=1}^{\infty} a_n$	Diverges if $\lim_{n\to\infty} a_n \neq 0$ or $\lim_{n\to\infty} a_n$ does not exist.	Inconclusive if $\lim_{n\to\infty} a_n = 0$.
p-series	$\sum_{n=1}^{\infty} \dfrac{1}{n^p}$	• Converges if $p > 1$ • Diverges if $p \leq 1$.	Useful for comparison tests if the nth term a_n of a series is *similar* to $1/n^p$.
Integral	$\sum_{n=1}^{\infty} a_n$, $a_n = f(n), n \geq 1$	• Converges if $\int_1^{\infty} f(x)dx$ converges • Diverges if $\int_1^{\infty} f(x)dx$ diverges The function f must be continuous, positive, and decreasing on $[1, \infty)$.	In order to use the Integral Test we need to be able to evaluate $\int_1^{\infty} f(x)dx$. The value of the integral is not the value of the series.
Comparison	$\sum_{n=1}^{\infty} a_n$, $\sum_{n=1}^{\infty} b_n$ $a_n > 0$, $b_n > 0$	• If $\sum_{n=1}^{\infty} b_n$ converges and $a_n \leq b_n$ for all n, then $\sum_{n=1}^{\infty} a_n$ converges. • If $\sum_{n=1}^{\infty} b_n$ diverges and $a_n \geq b_n$ for all n, then $\sum_{n=1}^{\infty} a_n$ diverges.	The comparison series $\sum b_n$ is usually a geometric series or a p-series. The series $\sum a_n$ is given, you supply $\sum b_n$.
Limit Comparison	$\sum_{n=1}^{\infty} a_n$, $\sum_{n=1}^{\infty} b_n$ $a_n > 0$, $b_n > 0$	If $\lim_{n\to\infty} \dfrac{a_n}{b_n} = c$, where $c > 0$ is a finite real number, then either both series converge or both diverge.	Find a suitable $\sum b_n$ whose convergence or divergence can be determined (e.g., a p-series).The series $\sum a_n$ is given, you supply $\sum b_n$.
Alternating Series	$\sum_{n=1}^{\infty} (-1)^n b_n$, $b_n > 0$, $n \geq 1$	Converges if • $b_{n+1} \leq b_n$ for all n, (i.e., $\{b_n\}$ is decreasing) • $\lim_{n\to\infty} b_n = 0$	Applicable only to an alternating series. If $b_n = f(n)$ and f is decreasing for $x \geq 1$, then $\{b_n\}$ is also decreasing.
Absolute Convergence	$\sum_{n=1}^{\infty} a_n$	Converges absolutely (and therefore converges) if $\sum_{n=1}^{\infty} \lvert a_n\rvert$ converges.	Useful for series that contain both positive and negative terms. Applies to arbitrary series.
Conditional Convergence	$\sum_{n=1}^{\infty} a_n$	If $\sum_{n=1}^{\infty} a_n$ is converges but $\sum_{n=1}^{\infty} \lvert a_n\rvert$ diverges.	
Ratio	$\sum_{n=1}^{\infty} a_n$	If $\lim_{n\to\infty} \left\lvert \dfrac{a_{n+1}}{a_n} \right\rvert = L$, then the series $\sum_{n=1}^{\infty} a_n$ is • absolutely convergent (and therefore convergent) if $L < 1$ • divergent if $L > 1$ or ∞	Inconclusive if $L = 1$. Useful if a_n involves factorials such as $n!$ or n^{th} powers. If $a_n > 0$ for every n, the absolute value sign may be disregarded. Test for series with nonzero terms.
Root	$\sum_{n=1}^{\infty} a_n$	If $\lim_{n\to\infty} \sqrt[n]{\lvert a_n\rvert} = L$, then the series $\sum_{n=1}^{\infty} a_n$ is • absolutely convergent (and therefore convergent) if $L < 1$ • divergent if $L > 1$ or ∞	Inconclusive if $L = 1$. Useful if a_n involves nth powers. If $a_n > 0$ for every n, then the absolute value sign may be disregarded.